

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1108

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture.
(BL3)

Assessment Tool Weightage: 40%

QUESTIONS: Any 4

Total Marks:40

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

*I Deepa lakshana.B (192471054) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

ANALYTICAL PROBLEM SOLVING

1. Use Case Diagram Problem

a) Actors

- Patient
- Doctor
- Administrator

b) Primary Use Cases

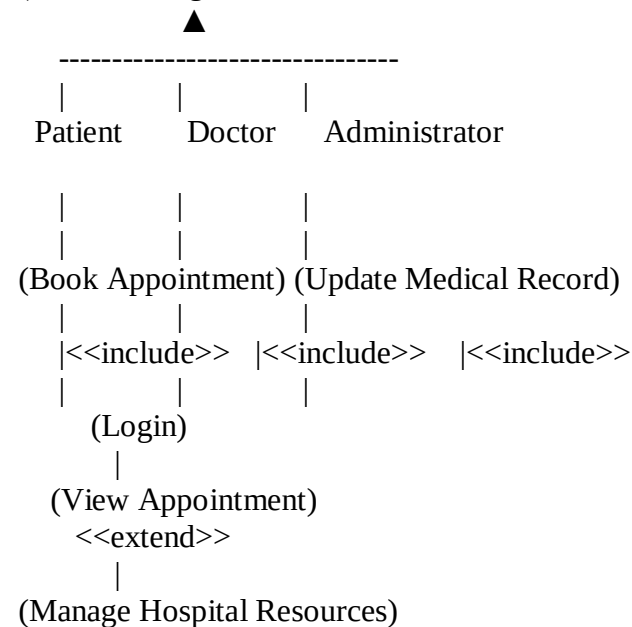
- Book Appointment
- View Appointment
- Update Medical Record
- Manage Hospital Resources
- Login

c) Relationships

- <<include>>: Book Appointment → Login
- <<include>>: Update Medical Record → Login
- <<include>>: Manage Hospital Resources → Login
- <<extend>>: View Appointment → Book Appointment
- **Generalization**: User → Patient, Doctor, Administrator

d) use case diagram

User



Benefits

- Clearly identifies **actors** and **system functions**.
- Shows relationships using **include**, **extend**, and **generalization**.
- Improves **requirement analysis, communication, and system design**.

2. Class Diagram Problem

a) Classes with Attributes and Methods

Book

- Attributes: BookID, Title, Author
- Methods: issueBook(), returnBook()

Member

- Attributes: MemberID, Name
- Methods: borrowBook(), returnBook()

Librarian

- Attributes: LibrarianID, Name
- Methods: addBook(), removeBook()

IssueRecord

- Attributes: IssueDate, ReturnDate
- Methods: createRecord(), closeRecord()

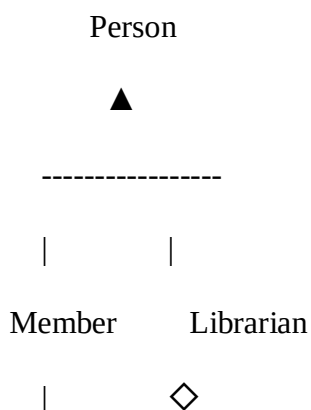
b) Relationships

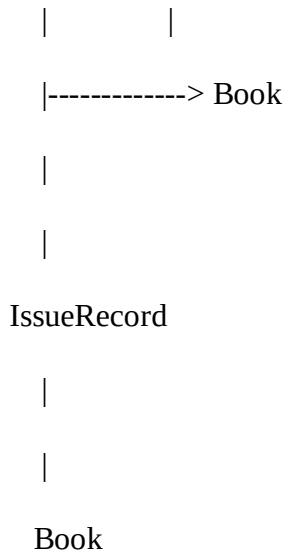
- **Association:** Member $\leftrightarrow$ Book (borrows books)
- **Aggregation:** Librarian $\diamond$ — Book (manages books)
- **Inheritance:** Person $\rightarrow$ Member, Librarian

c) Multiplicity

- **Member (1) $\rightarrow$ (0..*) IssueRecord**
- **Book (1) $\rightarrow$ (0..*) IssueRecord**
- **Librarian (1) $\rightarrow$ (0..*) Book**

d)class diagram





Benefits

- Clearly defines **classes, attributes, and methods**.
- Shows **association, aggregation, and inheritance** relationships.
- Improves **system design, reusability, and maintainability**.

3. Sequence Diagram Problem

a) Objects Involved

- Customer
- Food Ordering System
- Payment Gateway
- Restaurant

b) Message Flow

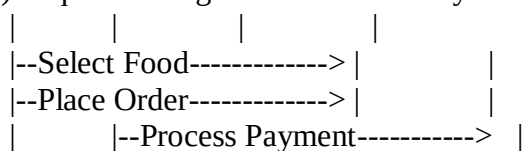
1. Customer → Select Food
2. Customer → Place Order
3. System → Payment Gateway (Process Payment)
4. Payment Gateway → System (Payment Success)
5. System → Restaurant (Send Order)
6. Restaurant → System (Confirm Order)
7. System → Customer (Order Confirmation)

c) Activation & Lifelines

- Each object has a **lifeline** (vertical dashed line).
- **Activation bars** appear while processing requests such as payment and order confirmation

d) Sequence Diagram

	Customer	System	Payment Gateway	Restaurant
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```

|      |<--Payment Success----- |
|      |--Send Order----->|
|      |<-----Order Confirmed-----|
|<--Order Confirmation-----|      |

```

Benefits

- Shows the **order of interactions** between objects.
 - Clearly represents **message flow, lifelines, and activations**.
 - Improves **system understanding and communication**.
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4. Activity Diagram Problem

a) Activities

- Submit Application
- Verify Documents
- Pay Admission Fee
- Confirm Enrollment

b) Decision Nodes

- Documents Verified?
 - **Yes** → Fee Payment
 - **No** → Reject Application
- Fee Paid?
 - **Yes** → Confirm Enrollment
 - **No** → Pending/Cancel Admission

c) Parallel Processing

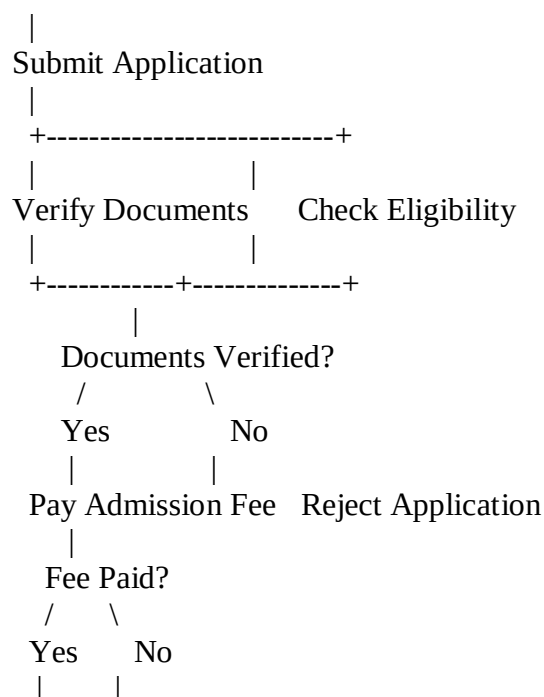
After application submission:

- Document Verification
- Eligibility Check

These two activities can run **in parallel** before proceeding to fee payment

d) Activity Diagram

Start



Confirm Pending/
Enrollment Cancel

|

End

Benefits

- Shows the complete **workflow** clearly.
- Includes **decision nodes** for approval/rejection.
- Uses **parallel processing** to improve efficiency.
- Makes the admission process **faster and easier to understand**.

5. State Diagram Problem

a) States

- Initial State
- Idle
- Product Selected
- Payment Received
- Product Dispensed
- Transaction Completed
- Final State

b) Events Triggering Transitions

- **Start** → Idle
- **Select Product** → Product Selected
- **Make Payment** → Payment Received
- **Dispense Product** → Product Dispensed
- **Complete Transaction** → Transaction Completed
- **Finish** → Final State

c) Initial and Final States

- **Initial State:** Start
- **Final State:** End

d) State Transition Diagram● (Start)

|

Idle

|

Select Product

|

Product Selected

|

Make Payment

|

Payment Received

|

Dispense Product

|

Product Dispensed

|

Complete Transaction

|

Transaction Completed

- ⊙(End) **Benefits**
- Clearly shows all **states** and **transitions**.
 - Represents **system behavior** based on events.
 - Improves **system design, testing, and maintenance**.
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6. Component Diagram Problem

a) Major Components

- User Management
- Course Management
- Assessment Module
- Content Delivery Module
- Notification Service

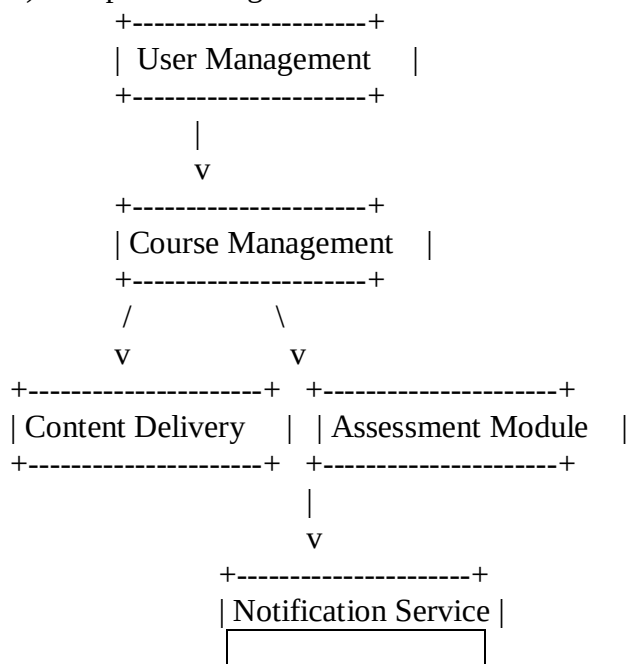
b) Interfaces

- **User Management** → Authentication/Login
- **Course Management** → Course Access
- **Assessment Module** → Exam & Results
- **Content Delivery Module** → Learning Content
- **Notification Service** → Email/SMS Notifications

c) Dependencies

- User Management → Course Management
- Course Management → Content Delivery Module
- Course Management → Assessment Module
- Assessment Module → Notification Service

d) Component Diagram



Benefits

- Identifies major **system components**.
 - Shows **interfaces and dependencies** between modules.
 - Improves **modularity, scalability, and maintainability** of the system.
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7. Deployment Diagram Problem

a) Hardware Nodes

- Roadside Sensor
- Cloud Server
- Traffic Authority Computer (Web Dashboard)

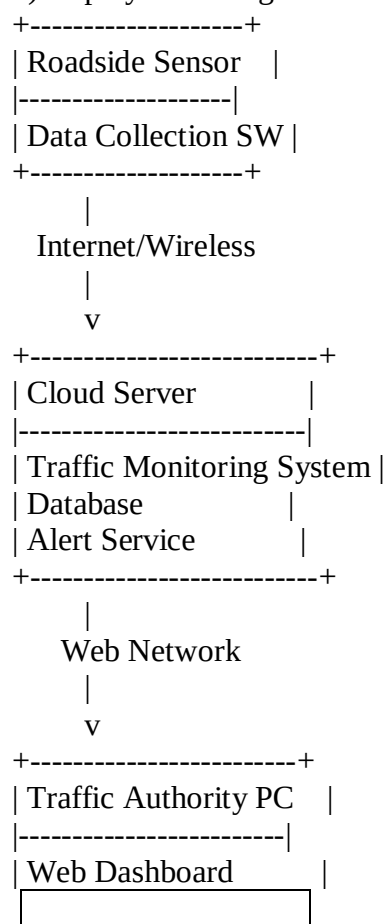
b) Software Components

- Roadside Sensor: Data Collection Software
- Cloud Server: Traffic Monitoring System, Database, Alert Service
- Traffic Authority Computer: Web Dashboard

c) Communication Links

- Roadside Sensor $\rightleftharpoons$ Cloud Server (Internet/Wireless)
- Cloud Server $\rightleftharpoons$ Traffic Authority Computer (Web Network)

d) Deployment Diagram



Benefits

- Shows **hardware nodes** and **software deployment**.
- Defines **communication links** between devices.
- Improves **system scalability, monitoring, and maintenance**.

RUBRICS FOR CALCULATION

ANALYTICAL PROBLEM SOLVING

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation